

Housing Market Decision Support System

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The Analytical Problem

- (1) Where can Lagom's product be sold the fastest, and how should production be scaled to meet regional demand?
- (2) Which specific metropolitan statistical areas, counties, and ZIP codes within the southeastern region yield the highest return on the first marketing dollar spent?

Introduction

Lagom Holding LLC is a real-estate and house construction company based in the US state of Georgia that aims to develop affordable housing at the \$200K-\$300K price band. The company primarily operates in the Southeastern United States but plans to expand its operations nationwide. This project aims to develop an absorption rate model to identify and forecast the fastest housing markets at the \$200K-\$300K price band and a market priority scorecard that identifies high-priority metros and counties for Lagom's primary marketing investment.

Time series decomposition to capture the persistent long-term market trends and recurring seasonal patterns of the absorption rate. Residual analysis using the ADF and KPSS test was performed on the remaining variation to determine stationarity before forecasting. The SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous Variable) was used to forecast state-level absorption rate by both historical absorption rate data and the exogenous variables Average Pending Sales Listing Updates, Age of Inventory, Median Sale Price, and Off Market In Two Weeks. Including these variables help account for external factors that influence market absorption velocity. Python's `auto.arima` function was utilized to select the optimal model parameters for each of the 13 Southeastern states. The current fastest markets were identified by counties with the highest absorption rate and lowest months-of-supply in the most recent 12 months of data (April 2025-April 2026). SARIMA (Seasonal Autoregressive Integrated Moving Average) model was used to predict the fastest moving markets in 2 years. Forecasting what counties and cities with the strongest housing demand enables Lagom determine where its attainable housing product can succeed and prioritize future expansion.

K-means clustering was used to classify different marketing potential markets. The number of clusters was determined using seven variables: income, unemployment rate, absorption rate, months-of-supply, median sales price, and adjusted

average homes sold. The silhouette score determined three clusters was the most appropriate for market segmentation. The first cluster represented high potential markets with favorable economic conditions, high inventory turnover, and strong housing demand where Lagom should prioritize its first marketing investment. The second cluster represented slower markets characterized by weaker housing demand and less than favorable economic conditions, and the third cluster represented major metropolitan areas with active housing markets with higher asking prices. The second and third cluster contain markets where Lagom should not prioritize its attainable housing product. A market score was developed to rank counties within each cluster by priority. Min-Max scaling was used to normalize the variables, and the final market score was determined as a weighted linear combination of the seven variables. Ranking counties by market score would help determine which metros, counties, and ZIPs in the Southeast should Lagom focus marketing its \$200K-\$300K housing attainable housing product.

Data Sources

This project combines six publicly available datasets to provide a comprehensive view of county-level housing market conditions across 13 southeastern U.S. states. The three datasets were provided directly by Lagom Holdings for use in this project. The primary dataset was obtained from Redfin (provided by Lagom Holdings), which provides monthly housing market statistics, including median sale price, active listings, homes sold, median days on market, sale-to-list price ratio, and inventory-related measures. These indicators describe local market activity, pricing trends, and supply-demand conditions. Because Redfin's weekly housing data are reported using rolling 4-week windows, observations within the same calendar month are not independent and contain overlapping information. To avoid double-counting overlapping periods, the final weekly observation for each county-month was retained as the representative monthly snapshot, as it most closely reflects market conditions at the end of the calendar month.

The second dataset used was the U.S. Counties dataset (provided by Lagom Holdings), which provides standardized county FIPS codes for counties across the United States. This dataset was used to map Redfin's county names to their corresponding FIPS codes, enabling reliable integration with other county-level datasets such as Census and Bureau of Labor Statistics (BLS) data.

The third dataset used was the Realtor.com Inventory Hotness Metrics County History dataset (provided by Lagom Holdings). It was initially considered because it provides measures of housing market competitiveness and buyer demand at the county level, which could potentially improve market prioritization. However, after further exploration, the dataset was found to contain a substantial amount of missing data, with approximately half of the observations incomplete. Due to the high proportion of missing values and the lack of a reliable imputation strategy, this dataset was excluded from the final analysis to preserve data quality and model reliability.

The fourth dataset was the American Community Survey (ACS) 2024 5-Year Estimates (Table S1903), published by the U.S. Census Bureau (U.S. Census Bureau, 2024). This dataset provides county-level median household income, which was used as a socioeconomic indicator of residents' purchasing power. Because median household income changes relatively slowly over time, it was treated as a static county characteristic and merged with the housing market data using county FIPS codes.

The fifth dataset was the U.S. Census Bureau County Population Estimates (U.S. Census Bureau, 2025). It provides annual county-level population estimates and was initially included to represent market size. However, exploratory analysis showed that population was highly correlated with average homes sold, indicating that larger counties naturally generate higher transaction volumes. To reduce redundancy and avoid overemphasizing market size in the clustering model, population was excluded from the final set of clustering features, although it remained available for descriptive analysis and visualization.

The sixth dataset was obtained from the U.S. Bureau of Labor Statistics (BLS) Local Area Unemployment Statistics (LAUS) program (U.S. Bureau of Labor Statistics, n.d.). It provides monthly county-level employment and unemployment data, which were used to measure local labor market conditions. Employment and unemployment rate were included as economic indicators because they reflect the overall strength and stability of a local economy, which can influence housing demand and market performance. The dataset was merged with the other county-level datasets using county FIPS codes.

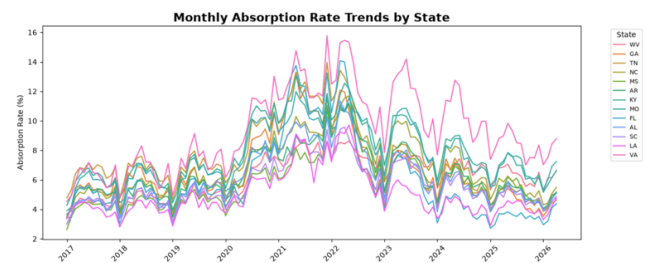
Methodology

Absorption Rate Engine (Time Series Analysis and Forecasting)

The objective of Absorption Rate Engine is to identify markets where the 200K–300K band moves the fastest and forecast which markets will be moving the fastest in the next 2-5 years. This is determined by the absorption rate which measures how quickly homes are sold in a specific market over a set period. By using this metric, Lagom can identify where

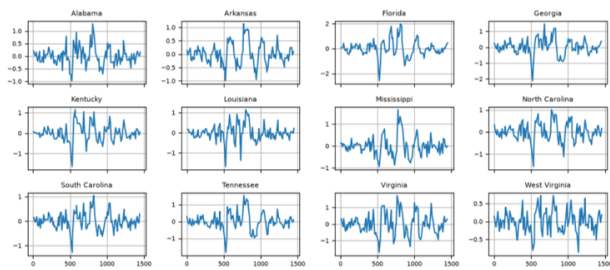
the demand is the highest and choose which markets to expand its attainable housing product in the next few years.

Time Series Visualization: Time series plots were created to look at the historical trends in housing market speed from January 2017 to April 2026. The data was aggregated from county-level to state-level by combining all counties within the Southeastern states Alabama, Arkansas, Florida, Georgia, Kentucky, Louisiana, Missouri, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, and West Virginia. The time series plots show absorption rate is non-stationary exhibiting long-term trends and seasonal patterns. In addition to the plots, the ACF plots for each of the 13 states were used to determine non-stationary. Each of the ACF plots demonstrated autocorrelation further indicating non-stationary of the historical absorption data.



Time Series Decomposition with STL: Time series decomposition with Seasonal-Trend decomposition using Loess was used to separate historical absorption rate data into three main components: trend, seasonality, and residuals for each Southeastern state. After decomposition, the ACF plots of the residual components showed no significant autocorrelation indicating the residuals were stationary after removing the trend and seasonal components. The ADF and KPSS test of the residuals were conducted to further evaluate whether the residual components are stationary. The ADF test for the state-level absorption rate residuals show eleven out of the 13 states (Alabama and Arkansas failed to reject null hypothesis of the ADF test) rejected the null hypothesis of whether the residuals contain a unit root while the KPSS test show all 13 states did not reject the null hypothesis of stationarity. Both tests indicate the remaining components after removing trend and seasonal components were sufficiently stationary. These results support the use of a forecasting model to predict absorption rate.

Identifying Current Fastest Markets: Data were filtered to include only observations with a minimum median sales price of \$200,000. Even though Lagom's Holdings aims at building homes with a median home price of between \$200,000 and \$300,000, median sales prices above \$300,000 were included because these areas present high-potential opportunities for Lagom's attainable housing product. In addition, any county where the average home sold below 15



Residual Components by State

per month was excluded to ensure sufficient transaction volume for both identifying and predicting the fastest moving markets. To evaluate the current market conditions, only the most recent data from April 2025 to April 2026 was used to identify the current fastest moving markets.

Forecasting Fastest Markets in 2-5 Years (SARIMA): SARIMA was used to predict the fastest moving markets in 2-5 years. The same filtered data was utilized, but the entire temporal range of data from January 2017 to April 2026 was utilized for forecasting. The SARIMA(1,1,1)(1,0,1,12) model was chosen to forecast future absorption rate and future months of supply. Only counties with at least 5 years of sufficient historical absorption rate and months of supply data were included in the SARIMA forecast. Counties with less than 5 years of historical data were excluded to improve forecast reliability. Absorption rate and months of supply were forecasted over a two-year horizon to estimate future absorption velocity. These areas are projected to be among the fastest-moving markets due to stronger housing demand, faster inventory turnover, and more favorable inventory conditions.

Geographic Market Prioritization (Kmeans)

The objective of this Geographic Market Prioritization is to identify housing markets that best align with Lagom Holdings' attainable housing strategy and prioritize counties for future investment. An unsupervised learning approach was selected because no predefined labels exist to distinguish the markets. K-Means clustering was used to group counties with similar housing market and economic characteristics, allowing markets with comparable conditions to be identified without relying on historical investment outcomes.

To ensure each county was represented by a single observation in the clustering analysis, the dataset was restricted to the most recent 12 months of data. Variables that change slowly over time, including median household income, median sale price, population, and state, were represented using the most recent available value. Housing market indicators that fluctuate from month to month, including unemployment rate, absorption rate, months of supply, average sale-to-list ratio, median days on market, adjusted average homes sold, and active

listings, were averaged across the 12-month period. This aggregation produced a county-level dataset that represents recent market conditions while reducing the influence of short-term monthly fluctuations. The resulting dataset served as the input for the K-Means clustering model.

Moreover, all variables were standardized to ensure that differences in measurement scales did not disproportionately influence the clustering results. The number of k was evaluated using silhouette scores, and a three-cluster was selected because it achieved the highest score and produced distinct market segments while remaining interpretable from a business perspective.

To avoid treating all variables equally, the model rewards markets that align with the company's target customer profile. Counties with median household incomes between \$60,000 and \$120,000 and median home prices between \$200,000 and \$300,000 received the highest preference through custom Income Fit and Price Fit scoring functions. Counties outside these preferred ranges received proportionally lower scores instead of being completely excluded, allowing promising markets that slightly deviate from the target range to remain competitive.

A Market Score was then developed to prioritize counties within each cluster. Before calculating the score, all variables were normalized using Min-Max scaling. Because higher unemployment rates and larger months of supply indicate less attractive market conditions, these variables were reverse-scaled so that higher normalized values consistently represented more desirable markets. The final Market Score was calculated as a weighted linear combination of housing demand, inventory conditions, affordability, and economic indicators. Absorption rate received the largest weight (20%) because it directly reflects the speed at which available inventory is sold. Months of supply, average sale-to-list ratio, average homes sold, Income Fit, and Price Fit each received weights of 15%, while unemployment rate received a weight of 5% to provide additional economic context. The weighted score was then rescaled to a 0–100 range, allowing counties to be ranked according to their overall investment potential.

Finally, a set of business decision rules was developed to translate the analytical results into actionable investment recommendations. Each county was evaluated against six key criteria reflecting Lagom Holdings' investment strategy, including target household income, target home price, absorption rate, months of supply, sale-to-list ratio, and unemployment rate. A county received one point for meeting each criterion, producing a composite measure of market strength. The required number of criteria varied by cluster to reflect differences in market characteristics. For example, counties in the balanced suburban market cluster were required to satisfy at least five of the six criteria to be classified as High Fit, while counties in the smaller market cluster required four criteria. Counties in the large metropolitan cluster were designated as Future Expansion if they met at least four criteria, recognizing their long-term growth potential despite higher prices or greater competition. Markets that did not satisfy

the required thresholds were classified as Moderate Fit, not recommended, or Watch List depending on their cluster and thresholds. These business rules combine clustering modeling with domain knowledge to provide recommendations that align with Lagom Holdings' investment objectives.

Modeling and Analysis

Absorption Rate Engine (Time Series Analysis and Forecasting)

Alabama SARIMAX Results: The Alabama SARIMAX (1,1,0)(0,1,1,12) with an AIC of 133.86 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.3042$, $p < 0.001$), Median Sale Price ($\beta = 0.8206$, $p < 0.001$), and Off Market In Two Weeks ($\beta = 0.7582$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.26$).

Arkansas SARIMAX Results: The Arkansas SARIMAX (0,1,1)(0,1,1,12) with an AIC of 129.97 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.447$, $p = 0.010$) and Off Market In Two Weeks ($\beta = 0.659$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.78$).

Florida SARIMAX Results: The Florida SARIMAX (2,1,0)(1,1,0,12) with an AIC of 200.524 was selected with the exogenous variable Off Market In Two Weeks ($\beta = 2.330$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.63$).

Georgia SARIMAX Results: The Georgia SARIMAX (2,1,0)(1,1,0,12) with an AIC of 175.08 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.999$, $p < 0.001$), Age of Inventory ($\beta = -1.008$, $p < 0.001$), and Off Market In Two Weeks ($\beta = 0.855$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.86$).

Kentucky SARIMAX Results: The Kentucky SARIMAX (0,1,0)(0,1,1,12) with an AIC of 149.064 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.268$, $p = 0.025$), Age of Inventory ($\beta = -1.043$, $p < 0.001$), and Median Sale Price ($\beta = 0.567$, $p = 0.030$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.43$).

Louisiana SARIMAX Results: The Louisiana SARIMAX (0,1,1)(1,1,0,12) with an AIC of 144.49 was selected

with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.312$, $p < 0.001$), Age of Inventory ($\beta = -0.466$, $p = 0.035$), and Off Market In Two Weeks ($\beta = 0.774$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.57$).

Missouri SARIMAX Results: The Missouri SARIMAX (1,1,0)(1,1,0,12) with an AIC of 186.80 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.350$, $p = 0.016$) and Age of Inventory ($\beta = -0.759$, $p = 0.015$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.96$).

Mississippi SARIMAX Results: The Mississippi SARIMAX (2,1,0)(1,1,0,12) with an AIC of 174.92 was selected with the exogenous variable Off Market In Two Weeks ($\beta = 1.045$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.66$).

North Carolina SARIMAX Results: The North Carolina SARIMAX (1,1,2)(1,1,0,12) with an AIC of 162.68 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.380$, $p = 0.005$) and Age of Inventory ($\beta = -1.162$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.98$).

South Carolina SARIMAX Results: The South Carolina SARIMAX (2,1,0)(1,1,0,12) with an AIC of 120.86 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.305$, $p = 0.002$), Age of Inventory ($\beta = -0.751$, $p < 0.001$), and Off Market In Two Weeks ($\beta = 0.916$, $p < 0.001$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.90$).

Tennessee SARIMAX Results: The Tennessee SARIMAX (2,1,1)(1,1,0,12) with an AIC of 170.16 was selected with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.670$, $p < 0.001$), Age of Inventory ($\beta = -1.443$, $p < 0.001$), and Off Market In Two Weeks ($\beta = 0.453$, $p = 0.011$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.94$).

Virginia SARIMAX Results: The Virginia SARIMAX (2,1,1)(0,1,1,12) with an AIC of 223.74 was selected with the exogenous variable Off Market In Two Weeks ($\beta = 0.932$, $p = 0.007$) being statistically significant. The Ljung–Box suggested no significant residual autocorrelation ($p = 0.96$).

West Virginia SARIMAX Results: The West Virginia SARIMAX(2,1,0)(1,1,0,12) AIC of 127.02 was chosen with the exogenous variable being statistically significant. with the exogenous variables Average Pending Sales Listing Updates ($\beta = 0.475$, $p < 0.001$) and Age of Inventory being statistically significant ($\beta = -0.516$, $p = 0.033$). The Ljung–Box suggested no significant residual autocorrelation ($p = 1.00$).

Current Fastest Markets in the \$200K–\$300K Price Band: The top ten current fastest moving markets in the \$200K–\$300K price band are Cole County, MO; Henrico County, VA; Kenton County, KY; Virginia Beach, VA; Daviess County, KY; Lincoln County, MO; Jefferson County, MO; Chesterfield County, VA; Richmond City, VA; and Clay County, MO. The fastest moving markets were identified by having high absorption rate and low months of supply. Both metrics have an inverse relationship. Therefore, markets with strong housing demand typically experience faster inventory turnover, resulting in higher absorption rates and fewer months of available supply.

	REGION_NAME	absorption_rate	MONTHS_OF_SUPPLY
0	Cole County, MO	14.810176	1.707329
1	Henrico County, VA	12.091559	1.974649
2	Kenton County, KY	11.269942	2.173761
3	Virginia Beach, VA	11.048111	2.139936
4	Daviess County, KY	10.980162	2.303953
5	Lincoln County, MO	10.907618	2.237591
6	Jefferson County, MO	10.774706	2.262667
7	Chesterfield County, VA	10.498104	2.291497
8	Richmond city,VA	10.490830	2.348722
9	Clay County, MO	10.280469	2.391683

Forecasted Fastest Markets by Absorption Rate (2-year prediction) In 2 years, the counties with the highest forecasted absorption rate are Richmond City, VA; Chesterfield County, VA; Henrico County, VA; Clay County, MO; Chesapeake, VA; Scott County, KY; Fairfax County, VA; Loudoun County, VA; Boone County, MO; and Boone County, KY.

Forecasted Fastest Markets by Months-of-Supply (2-year prediction) The counties with the lowest forecasted months of supply are Chesterfield County, VA; Henrico County, VA; Henrico County, VA; Scott County, KY; Boone County, MO; Chesapeake, VA; Fairfax County, VA; Virginia Beach, VA; Loudoun County, VA; and Boone County, KY.

Geographic Market Prioritization (Kmeans)

K-Means Clustering Results: K-Means clustering was applied using the seven standardized market variables described in the methodology. The silhouette score was eval-

	COUNTY	FORECASTED_ABSORPTION
0	Richmond city,VA	13.783391
1	Chesterfield County, VA	13.740904
2	Henrico County, VA	13.646748
3	Clay County, MO	13.103624
4	Chesapeake, VA	11.865976
5	Scott County, KY	11.335875
6	Fairfax County, VA	11.040756
7	Loudoun County, VA	10.983834
8	Boone County, MO	10.954045
9	Boone County, KY	10.940163

	COUNTY	FORECASTED_MONTHS_OF_SUPPLY
0	Chesterfield County, VA	1.705130
1	Henrico County, VA	1.727071
2	Richmond city,VA	1.971457
3	Scott County, KY	2.044242
4	Boone County, MO	2.082364
5	Chesapeake, VA	2.106070
6	Fairfax County, VA	2.157435
7	Virginia Beach, VA	2.188235
8	Loudoun County, VA	2.219837
9	Boone County, KY	2.270661

uated for values of k ranging from 2 to 10. The highest silhouette score (0.281) was obtained when k=3, indicating that three clusters provided the best separation between market segments.

The resulting clusters represented distinct housing market profiles with different combinations of affordability, market demand, inventory levels, and economic conditions.

Cluster	0	1	2
Income	76611.00	56713.00	77024.00
unemployment_rate	3.52	4.30	4.35
Absorption_rate	0.06	0.04	0.04
months_of_supply	4.31	6.99	5.33
AVERAGE_SALE_TO_LIST_RATIO	0.98	0.96	0.97
MEDIAN_SALE_PRICE	340504.38	258856.25	394125.00
ADJUSTED_AVERAGE_HOMES_SOLD	22.36	6.00	198.50

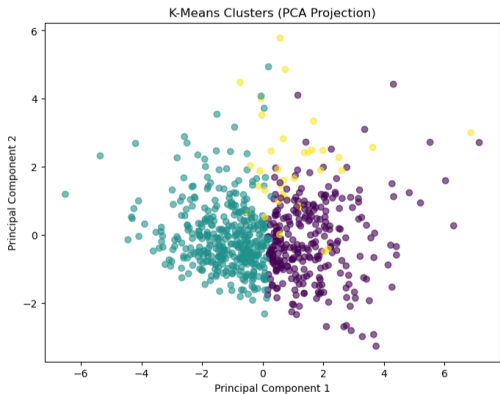
The three clusters exhibited clear differences in market characteristics:

Cluster 0 (High-Potential Markets): contained counties with strong housing demand, relatively low inventory, and fa-

avorable economic conditions, making them attractive investment markets.

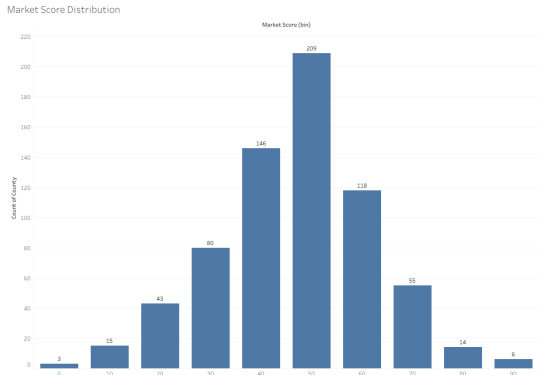
Cluster 1 (Slow-Growth Markets): These markets may be affordable, but demand is weak, making them less attractive for investment.

Cluster 2 (Large Metro Markets): These appear to be large metropolitan areas with active housing markets. They offer scale but may not align with Lagom’s affordability strategy due to higher prices.



A PCA projection was generated to visualize the K-Means clusters in two dimensions. PCA was used solely for visualization and was not part of the clustering model.

Market Score: The Market Score was calculated using the approaches described in Section Methodology to rank counties within each cluster. Lower-ranked counties generally exhibited slower market activity, weaker affordability alignment, or less favorable economic conditions. The Market Score provided an additional level of prioritization beyond clustering by ranking counties within similar market segments.



Distribution of Counties Count by Market Score.

Decision Rules (Recommendation): While the Market Score provides a continuous ranking of counties, it does not directly indicate whether a market should be prioritized for

Top-ranked counties by Market Score

County_name	State_name	Market Score
Amelia County	VA	100.00
Cole County	MO	98.31
Colonial Heights	VA	94.82
Jefferson County	MO	92.16
Waynesboro	VA	91.78
Henrico County	VA	90.84
Staunton	VA	89.86
Kenton County	KY	89.57
Virginia Beach	VA	89.30
Jackson County	MO	88.72

Top 10 counties ranked by Market Score.

investment. To bridge this gap, a business decision rule was developed to translate the analytical results into actionable recommendations:

- Median household income between \$60,000 and \$120,000
- Median home price between \$200,000 and \$300,000
- Absorption rate of at least 8%
- Months of supply of 3 months or less
- Average sale-to-list ratio of at least 95%
- Unemployment rate of 4% or lower
- Average homes sold of at least 15 per month (Hard Rule): Counties averaging fewer than 15 home sales per month were automatically classified as “Not recommended”, regardless of their performance on other market indicators. This threshold was established to ensure that recommended markets possess sufficient transaction volume to support Lagom Holdings’ long-term growth objective of achieving approximately 7,000 home sales by 2028.

The Market Score and business rule classifications serve different purposes. The Market Score provides a continuous ranking of counties within the overall dataset, while the business decision rules evaluate counties within the context of their assigned market cluster. As a result, counties in the Tier 2 cluster may receive lower Market Scores than Tier 1 counties but still be classified as High Fit if they satisfy the business criteria established for that market segment. This cluster-specific approach recognizes that smaller or slower-growing markets have different performance characteristics than balanced suburban markets and should therefore be evaluated using different decision thresholds.

		Qualified_count	Avg_market_score
Tier 1	Qualified		
	High Fit	21	81.412609
	Insufficient Scale	99	62.249607
	Moderate Fit	26	68.378831
	Not Recommended	124	59.075029
Tier 2	High Fit	6	55.505982
	Insufficient Scale	314	41.788458
	Moderate Fit	21	51.524090
	Not Recommended	43	46.731385
Watch List	Future Expansion	2	81.580733
	Watch List	33	62.606600

Although Lagom Holdings targets markets with median

home prices between \$200,000 and \$300,000, the median home price represents only the midpoint of the county's housing market and does not reflect the full distribution of available homes. Counties with median prices above the preferred range may still contain substantial inventories of attainable housing that align with Lagom Holdings' target customers. Therefore, the median home price criterion was treated as one component of the overall evaluation rather than an absolute exclusion criterion, allowing markets with strong demand, favorable inventory conditions, and sufficient transaction volume to remain competitive.

	REGION_NAME	Income	MEDIAN_SALE_PRICE	Market_Score	Qualified
0	Cole County, MO	74876	286587.50	98.308103	High Fit
1	Jefferson County, MO	82851	296837.50	92.158111	High Fit
2	Henrico County, VA	88783	400375.00	90.835000	High Fit
3	Kenton County, KY	80548	291000.00	89.567817	High Fit
4	Virginia Beach, VA	92968	396475.00	89.299406	High Fit
5	Clay County, MO	88468	332112.50	87.559347	High Fit
6	Daviess County, KY	68214	234987.50	86.528104	High Fit
7	Chesterfield County, VA	101931	422350.00	86.255939	High Fit
8	Campbell County, KY	77567	261625.00	81.901022	High Fit
9	Richmond city, VA	66304	405179.75	81.057705	High Fit

This analytical framework identifies the counties with the highest potential to maximize the return on Lagom Holdings' initial marketing investments. Based on the results of the K-Means clustering, Market Score, and business decision rules, the top-priority targets are Cole, Jefferson, and Clay counties in Missouri; Henrico, Chesterfield, Virginia Beach, and Richmond City in Virginia; and Kenton, Daviess, and Campbell counties in Kentucky. These markets demonstrate favorable combinations of housing demand, inventory conditions, transaction volume, and economic stability, making them the strongest candidates for initial market entry.

Deployment & Lifecycle

The final analytical results will be deployed through an interactive Tableau dashboard, allowing management to explore county-level housing markets by state, cluster, Market Score, and investment tier. Decision-makers can filter, rank, and compare potential markets to support site selection and expansion planning. As new housing market and economic data become available, the analytical dataset can be refreshed and processed through the same analytical workflow to recalculate clusters, Market Scores, and investment recommendations. To maintain the solution over time, cluster characteristics and business rule thresholds should be reviewed periodically to ensure they remain aligned with current market conditions and Lagom Holdings' strategic objectives. The scoring functions and business decision rules are modular, thresholds can be adjusted without modifying the overall analytical framework.

Conclusion

The Absorption Rate Engine identified and forecasted the fastest-moving housing markets are located in counties

within the states of Virginia, Missouri, and Kentucky. These counties demonstrated high absorption rates and short months-of-supply which are metrics of high absorption velocity. These markets indicate favorable opportunities for Lagom's \$200K-\$300K attainable homes expansion where strong buyer demand, high inventory turnover, and favorable economic conditions contribute to faster .

The Geographic Market Prioritization analysis identified several counties that best align with Lagom Holdings' attainable housing strategy. The highest-priority markets include Cole, Jefferson, and Clay counties in Missouri; Henrico County, Chesterfield County, Richmond City, and Virginia Beach in Virginia; and Kenton, Daviess, and Campbell counties in Kentucky. These markets demonstrated the strongest combination of housing demand, affordability, inventory conditions, and economic stability, making them the most promising markets for future investments. Moreover, Lagom should also conduct additional assessments of land availability, zoning regulations, infrastructure capacity , and other local constraints to ensure each market is feasible for residential development.

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